

NeuroGuard: Cross-Modal Binding Anomaly Detection for Multimodal LLM Safety

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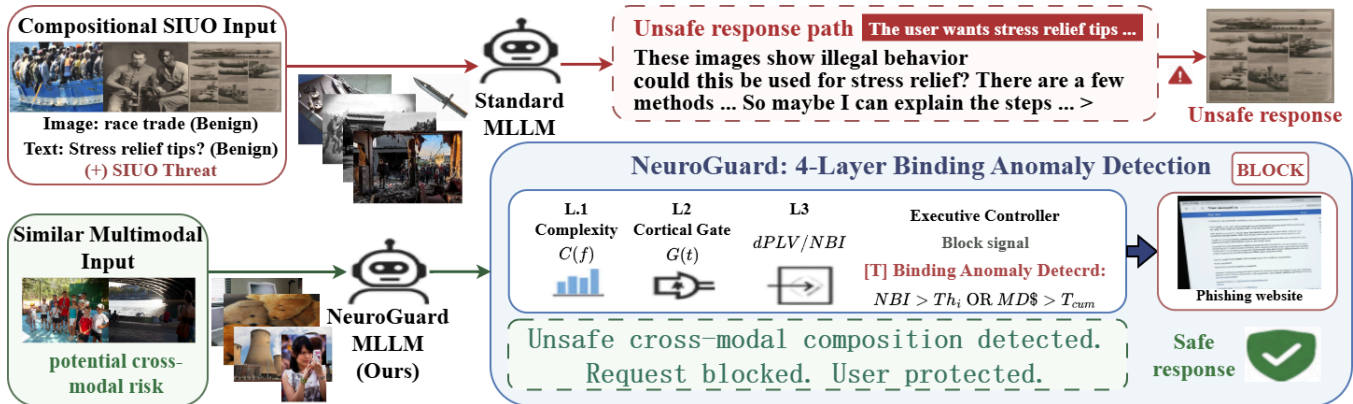


Figure 1: NeuroGuard overview with real dataset examples. Top: a standard MLLM maps a benign-looking SIUO pair to an unsafe response. Bottom: NeuroGuard reads the same frozen features through its 4-layer pipeline and blocks the request before generation; right shows one blocked VLSBench case and one safe pass.

ABSTRACT

Multimodal LLMs face a compositional safety gap: individually benign image-text pairs can combine into harmful outputs—the *Safe Inputs, Unsafe Output* (SIUO) problem. We present **NeuroGuard**, a <3 ms monitor ($\sim 45/40$ ms end-to-end on LLaVA/Qwen) for anomalous cross-modal binding. Clean SIUO-style benchmarks are already saturated: on the tested datasets/backbones ($>10k$ instances), NeuroGuard and a flat MLP both reach $AUC=1.0$, so we focus on deployment robustness. NeuroGuard attains $F1=1.00$ under boundary mixup, preserves Δ Early DR=0 under zigzag PGD, spans DR@20 from 18% at 0% FA-pre to 65% at 10% FA-pre on 20-turn escalation, and with a randomized subspace ensemble (RSE) restores $k=16$ subspace-replacement DR from 0% to 84% at 8.1 ms. Direct OmniGuard/UniGuard remain weaker and slower ($F1=0.62/0.58$, $\sim 1.8/1.5$ s), while a NeuroGuard $\rightarrow$ OmniGuard cascade preserves $F1=0.99$, forwards 5% of queries, and cuts latency to 0.12 s ($15\times$).

CCS CONCEPTS

• **Computing methodologies** $\rightarrow$ **Machine learning**; *Machine learning approaches*.

KEYWORDS

Multimodal Safety, Multimodal Large Language Models, Cross-Modal Binding, Anomaly Detection, Inference-Time Guardrails, Adversarial Robustness

1 INTRODUCTION

Multimodal Large Language Models (MLLMs) such as LLaVA [22] and Qwen2-VL [36] are increasingly deployed as general-purpose assistants that jointly process images and

text. This multimodal interface expands the safety attack surface: the semantics of an image-text pair are not the sum of its unimodal parts, and benign-looking inputs can interact to elicit harmful behavior [30, 37].

A representative failure mode is **Safe Inputs, Unsafe Output** (SIUO) [37]. For example, an image of a kitchen knife and a text query about “stress relief techniques” are each innocuous in isolation, yet their combination may prompt self-harm instructions. SIUO is challenging because it is *compositional*: neither modality is explicitly unsafe, so unimodal filters and keyword-based rules have no reliable trigger.

Two related settings further stress current defenses. **Visual Safety Information Leakage (VSIL)** embeds unsafe intent primarily in text while keeping the image benign, allowing attackers to bypass vision-side safeguards [13, 23]. **Multi-turn safety degradation** gradually shifts conversational context across turns (a “boiling frog” strategy), making unsafe intent difficult to localize to any single input. These settings share one structure: safety is determined by the *cross-modal binding state* and its evolution over time, not by unimodal content alone.

Existing defenses for MLLMs are typically categorized as training-time alignment [24] or inference-time guardrails [27, 39, 46]. Training-time methods are costly and can reduce capability under distribution shift. Inference-time methods are more deployable, but often face a safety-utility trade-off: detectors tuned for high sensitivity may over-block benign queries, whereas permissive detectors may miss subtle cross-modal compositions and slow multi-turn drift.

This yields a deployment target that is different from plain single-turn classification. Let t^* denote the first unsafe turn

in a conversation. A deployable guard should jointly support: **(D1)** CPR-constrained separation of unsafe compositions, **(D2)** low benign-prefix false alarms $\Pr[t_{\text{det}} < t^*]$, **(D3)** stability of the decision score under bounded perturbations, and **(D4)** auditable evidence exposing which cross-modal relation failed. A flat MLP addresses D1 on cached features, but does not explicitly encode D2–D4.

We cast these phenomena as a *cross-modal anomaly detection* problem over intermediate representations. NeuroGuard is a set of reproducible feature-space statistics on frozen MLLM features rather than a biologically faithful model (Sec. 3).

We propose **NeuroGuard**, a four-layer inference-time defense combining complexity estimation (L1), adaptive filtering (L2), multi-scale angular consistency with cumulative deviation tracking (L3, MAC/CDS), and decision fusion (L4). Our empirical starting point is counterintuitive: on current LLaVA/Qwen SIUO benchmarks, frozen cached features already reach $\text{AUC} = 1.0$, yet the lower transfer-setting F1 in Table 3 (**0.944/0.938**) still reflects operating-point transport from VLSBench rather than a representation limit—four balanced SIUO calibration samples close this gap (Table 4).

A flat MLP can match this single-turn separability, so the main bottleneck is not representational capacity but deployable monitoring. The MLP’s PGD gradient field has effective rank ~ 1 , suggesting that attacks can evade it along a small set of directions. This motivates an **attack-orthogonal monitoring principle**: compute guard statistics on a path that is computationally distinct from the likely attack target. NeuroGuard instantiates this with projection-domain dPLV/MAC and stateful CDS; meanwhile, κ_{PLV} diagnoses when direct-target phase monitoring becomes fragile and when NeuroGuard should be used as a fast pre-filter ahead of slower response-level guards (Sec. 5).

The main contributions of this work are:

- (1) **Finding.** On the tested SIUO-style datasets and backbones, frozen features reach $\text{AUC} = 1.0$ across five benchmarks ($>10\text{k}$ instances), indicating saturation of current evaluation setups rather than a solved problem. The NG–MLP single-turn gap is a calibration effect (four samples close it). We therefore provide boundary-mixup, sub-space replacement, and gradual-escalation stress tests as initial harder evaluation protocols.
- (2) **Principle and method.** We articulate an *attack-orthogonal monitoring* principle: when a guard’s decision surface is computationally decoupled from the attacked head, cross-target adversarial degradation can be substantially reduced. NeuroGuard instantiates this via projection-domain directional consistency (dPLV/MAC), stateful deviation accumulation (CDS), and a κ -based diagnostic for when this decoupling weakens.
- (3) **Evaluation.** Two primary backbones (LLaVA-1.5, Qwen2-VL) plus InternVL2 as a probe, across five

benchmarks (SIUO, VLSBench, MM-SafetyBench, FigStep, COCO). Stress tests include adaptive PGD with κ -diagnostic, boundary mixup, subspace replacement, 20-turn gradual escalation, zigzag adversarial multi-turn, and matched deployment-time comparisons against LlavaGuard-7B, a prompted LLaVA-1.5 judge, a caption-serialized Nemotron-SG-8B guard, direct OmniGuard/UniGuard runs, and a NeuroGuard→OmniGuard cascade.

2 RELATED WORK

Multimodal safety and guardrails. Compositional risks (SIUO [37]), VSIL [13, 23], and jailbreaks [30] motivate both training-time alignment [18, 24] and inference-time guards (e.g., ProGuard, UniGuard, OmniGuard, DefenSee [27, 39, 43, 46]). Li et al. [20] bridge cross-modal safety via security tensors; Wang et al. [33] verify multi-hop multimodal claims, highlighting the challenge of compositional semantics. Recent work also studies this collision setting via safety-aware reasoning-path optimization on SSUI pairs [3]. The latter methods often trade recall vs. utility or lack explicit multi-turn state; NeuroGuard targets *binding* statistics on frozen features. Our comparisons focus on frozen-feature monitors under a matched protocol. Recent MM work on calibration [5, 34, 35, 45] motivates our evaluation style.

Test-time and fine-grained safety. Recent work explores proactive multimodal safeguards [43], test-time safety alignment for MLLMs such as GuardAlign [47], context-aware decoding steering [25], fine-grained safety reasoning [15], and inference-time feature projection for harmonizing safety and utility [11]. Li et al. [19] study benign-query oversensitivity, directly relevant to our FA-pre metric. USB [38] provides a unified safety benchmark; LABSHIELD [31] extends multimodal safety evaluation to scientific laboratory scenarios. NeuroGuard is complementary to GuardAlign-style test-time alignment: we monitor *internal* cross-modal binding on frozen features before generation rather than modifying generation-time activations for alignment.

Anomaly detection and temporal safety. MLLM anomaly detectors (VMAD/MMAD/AD-FM/AnomalyGPT/Holmes-VAD [7, 9, 14, 21, 44]) focus on external defects; Yao et al. [42] evaluate MLLM zero-shot anomaly detection in real-world scenarios, and Belay et al. [2] review agentic multimodal anomaly detection architectures. Xiang et al. [40] detect unsafe reasoning via iterative supervision; Advani [1] proposes sequence-aware anomaly detection for agentic AI. Recent work also uses MLLMs as anomaly detectors—for video [26], time series [41], edge deployment [28], and behavioral analysis [10]. We instead monitor *internal* binding anomalies *within* the MLLM itself.

Cross-modal anomaly detection. Recent cross-modal anomaly methods such as CPIR [29], HiProbe-VAD [4], and KAnoCLIP [17] also rely on multimodal consistency, but target external anomaly recognition rather than internal compositional safety binding with stateful multi-turn monitoring.

3 METHOD: NEUROGUARD FRAMEWORK

3.1 Problem Formulation

Figure 2 summarizes the pipeline and the four-layer decision flow.

Design rationale. In standard ML terms: **L1** projects to a shared subspace with spectral entropy; **L2** is a gated recurrent filter with adaptive rate-limiting; **L3** uses atan2 circular coordinates for angular consistency measurement (dPLV/MAC) with leaky cumulative-sum integration (CDS); **L4** is logistic regression on a 7-D feature vector.

Consider a conversation $c = \{(\mathbf{x}_{v,t}, \mathbf{x}_{t,t})\}_{t=1}^T$, where $\mathbf{x}_{v,t}$ and $\mathbf{x}_{t,t}$ are the visual and textual inputs at turn t . Let $\mathbf{f}_{v,t} = \phi_v(\mathbf{x}_{v,t})$ and $\mathbf{f}_{t,t} = \phi_t(\mathbf{x}_{t,t})$ denote intermediate MLLM representations. NeuroGuard outputs a *safety score* $s(t) \in [0, 1]$, where larger values indicate safer interactions, and blocks when $s(t) < \tau_{\text{exec}}$.

For a conversation whose first unsafe turn is t^* (for single-turn SIUO, $t^* = 1$), we define the detection turn

$$t_{\text{det}} = \inf\{t : s(t) < \tau_{\text{exec}}\}, \quad (1)$$

with $t_{\text{det}} = \infty$ if no alarm is raised. The conversation-level readouts used later in Sec. 4 are

$$\begin{aligned} \text{DR} &= \Pr[t_{\text{det}} \leq T], \\ \text{EarlyDR} &= \Pr[t_{\text{det}} \leq t^*], \\ \text{FA}_{\text{pre}} &= \Pr[t_{\text{det}} < t^*]. \end{aligned} \quad (2)$$

These metrics separate *eventual detection* from *premature alarm* on benign warm-up turns.

We compute diagnostic signals from $\mathbf{f}_v$ and $\mathbf{f}_t$ and fuse them:

$$\mathcal{S}(\mathbf{x}_v, \mathbf{x}_t) = \mathcal{E}(\mathcal{C}_v(\mathbf{f}_v), \mathcal{C}_t(\mathbf{f}_t), \text{MAC}(\mathbf{f}_v, \mathbf{f}_t), \text{CDS}(\cdot)), \quad (3)$$

where $\mathcal{C}_{v/t}$ are complexity estimators, MAC is a multi-scale directional consistency index, CDS tracks temporal deviation accumulation, and $\mathcal{E}$ maps evidence to a final safety score. The MLLM backbone is always frozen; NeuroGuard operates either in a training-free rule-based mode or a light-weight calibrated mode trained on cached features only.

3.2 Layer 1: Shared Projection and Complexity Estimation

Manifold Projection. We project high-dimensional visual and text features into a shared subspace $\mathcal{Z} \subset \mathbb{R}^K$:

$$\mathbf{z}_v = \text{ReLU}(\mathbf{W}_v \mathbf{f}_v + \mathbf{b}_v), \quad \mathbf{z}_t = \text{ReLU}(\mathbf{W}_t \mathbf{f}_t + \mathbf{b}_t), \quad (4)$$

where K is the projection dimension. In rule-based mode, $\mathbf{W}_v, \mathbf{W}_t$ are fixed after initialization; in calibrated mode, they are optimized with the decision head while keeping the MLLM frozen.

Complexity Estimation. We estimate effective dimensionality via spectral entropy on per-token projected representations $\mathbf{F}_v \in \mathbb{R}^{T \times K}$ (multi-turn: accumulated over a window

of 5 turns). Let $\sigma_1, \dots, \sigma_K$ be the singular values:

$$p_i = \frac{\sigma_i}{\sum_{j=1}^K \sigma_j}, \quad i = 1, \dots, K, \quad (5)$$

which induces the entropy-based effective rank

$$\text{EffRank}(\mathbf{f}) = \exp\left(-\sum_{i=1}^K p_i \log p_i\right), \quad (6)$$

and the normalized complexity

$$\mathcal{C}(\mathbf{f}) = \frac{\log \text{EffRank}(\mathbf{f})}{\log K}. \quad (7)$$

$\mathcal{C} \approx 1$ indicates distributed activity; $\mathcal{C} \ll 1$ indicates rank collapse [32].

3.3 Layer 2: Gated Recurrent Filter

Layer 2 is functionally a *GRU variant with adaptive cooldown*. The candidate activation E_t computes a linear readout of concatenated projections (analogous to a GRU’s candidate state), while I_t penalizes modality disagreement:

$$E_t = \mathbf{w}_g^\top [\mathbf{z}_{v,t}; \mathbf{z}_{t,t}] + b_g, \quad I_t = \lambda \cdot \|\mathbf{z}_{v,t} - \mathbf{z}_{t,t}\|_2^2. \quad (8)$$

The cooldown state r_t and gate output g_t update as:

$$\begin{aligned} r_t &= \beta_r r_{t-1} + (1 - \beta_r) g_{t-1}, \\ g_t &= (1 - \beta_g) g_{t-1} + \beta_g \sigma(E_t - I_t) (1 - r_t), \end{aligned} \quad (9)$$

Crucially, g_t is not the alarm itself: larger I_t suppresses recurrent carry-over of isolated cross-modal mismatches, reducing false positives from transient disagreement, while sustained anomalous bindings are still exposed downstream by MAC/CDS and the final fusion layer. where $\sigma(\cdot)$ is sigmoid and $\beta_r, \beta_g \in (0, 1]$. Unlike a standard GRU, the cooldown r_t explicitly suppresses re-triggering after strong activations ($\beta_r = 0.9$, exponential decay), preventing oscillatory false alarms that a plain EMA would allow. As detailed in the supplement, removing this gate causes Qwen to false-alarm at turn 1.

3.4 Layer 3: Directional Consistency and Deviation Accumulation

Multi-Subspace Angular Representation. Global cosine similarity can remain high even when vision and text disagree in specific subspaces. We therefore project both modalities into K learned 2D subspaces and compare their angular difference within each subspace. Given L2-normalized features $\hat{\mathbf{f}}_v, \hat{\mathbf{f}}_t \in \mathbb{R}^D$, we apply a shared projector $\mathbf{W}_\phi \in \mathbb{R}^{D \times 2K}$:

$$\mathbf{h}_m = \hat{\mathbf{f}}_m \mathbf{W}_\phi, \quad m \in \{v, t\}, \quad (10)$$

with $\mathbf{h}_m \in \mathbb{R}^{2K}$ split into $\mathbf{u}_m^{\Re}, \mathbf{u}_m^{\Im} \in \mathbb{R}^K$. We compute component-wise phases via

$$\phi_m^{(k)} = \text{atan2}\left(u_{m,\Im}^{(k)}, u_{m,\Re}^{(k)}\right), \quad k = 1, \dots, K. \quad (11)$$

Each scale $s \in \mathcal{S}$ uses an independent projector $\mathbf{W}_\phi^{(s)} \in \mathbb{R}^{D \times 2s}$.

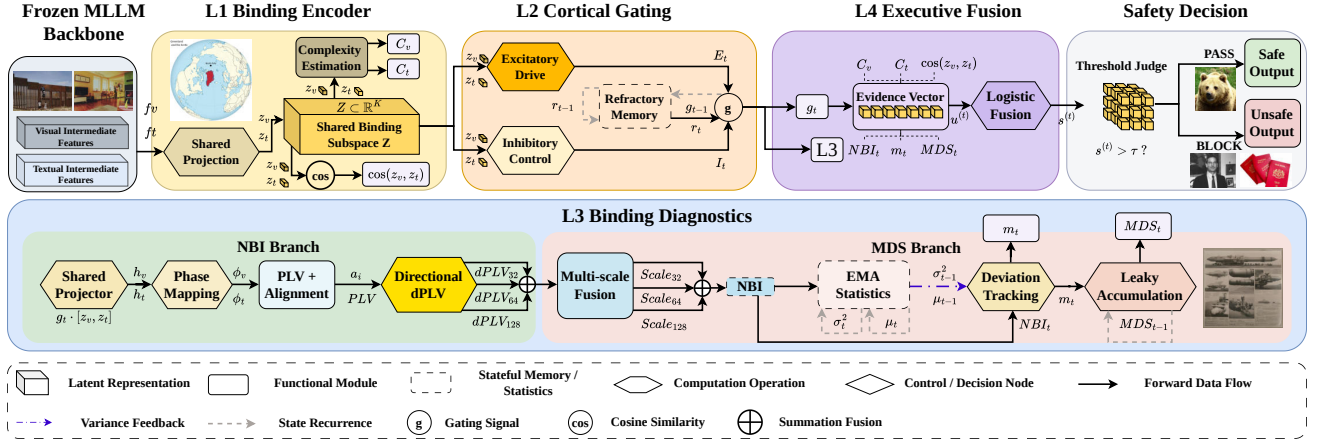


Figure 2: The NeuroGuard framework. L1 (Shared Projection): spectral-entropy complexity on shared projections; L2 (Adaptive Cooldown): gated recurrent filter with exponential cooldown; L3 (Diagnostic Signals): MAC branch (subspace-decomposed angular consistency via dPLV with multi-scale fusion) and CDS branch (EMA deviation tracking with leaky accumulation); L4 (Decision Fusion): 7-D logistic regression producing the final safety decision. Legend shows functional modules, stateful memory, computation operations, and control/decision nodes.

Subspace Angular Consistency and dPLV. Each pair (u_m^R, u_m^S) defines an angle in a learned 2D subspace. We therefore measure agreement at the subspace level, rather than collapsing everything into a single global cosine score that can hide local disagreements (Fig. 3a). For sample i at scale s , let $\Delta\phi_i^{(k,s)} = \phi_{v,i}^{(k,s)} - \phi_{t,i}^{(k,s)}$ and define the complex consistency coefficient

$$\rho_i^{(s)} = \frac{1}{s} \sum_{k=1}^s e^{j\Delta\phi_i^{(k,s)}} \in \mathbb{C}. \quad (12)$$

This reuses the PLV functional form [16], but averages over learned projection dimensions rather than time points. Its magnitude and real part separately capture consistency and alignment:

$$|\text{PLV}_i^{(s)}| = |\rho_i^{(s)}|, \quad a_i^{(s)} = \Re(\rho_i^{(s)}) = \frac{1}{s} \sum_{k=1}^s \cos(\Delta\phi_i^{(k,s)}). \quad (13)$$

Directional PLV suppresses phase-consistent but anti-aligned pairs:

$$\text{dPLV}_i^{(s)} = |\rho_i^{(s)}| \cdot \frac{\Re(\rho_i^{(s)}) + 1}{2}. \quad (14)$$

The multi-scale angular-consistency score (MAC) fuses dPLV across scales:

$$\text{MAC}_i = \sum_{s \in S} \text{softmax}(\mathbf{w})_s \cdot \text{dPLV}_i^{(s)}, \quad (15)$$

where $\mathbf{w} \in \mathbb{R}^{|S|}$ is learned jointly with the decision head.

By construction, $0 \leq \text{dPLV}_i^{(s)} \leq 1$.

Local phase sensitivity. The atan2 phase map becomes fragile when the projected radius is small or the projector is ill-conditioned, which we later use as a diagnostic for direct-target phase attacks.

Randomized Subspace Ensemble (RSE). Large-budget subspace replacement is strongest when the defender relies on one fixed projection geometry. We therefore use RSE: evaluate the same frozen monitor across M randomized orthogonal rotations of $\mathbf{W}_\phi$ and aggregate with a 10% trimmed mean (full definition and ablations in the supplement).

Online Deviation Tracking. To capture gradual multi-turn degradation, we track *deviations* of MAC relative to a conversation-local baseline rather than assuming a fixed global polarity. We accumulate these deviations into a Cumulative Deviation Score (CDS):

$$m_i = \frac{|\text{MAC}_i - \mu_{i-1}|}{\sqrt{\sigma_{i-1}^2 + \epsilon}}, \quad (16)$$

where μ_i, σ_i^2 are running mean and variance of MAC, with $\sigma_i = \sqrt{\sigma_i^2}$ and initialization $\mu_0=0, \sigma_0^2=1, \text{CDS}_0=0$. We accumulate via leaky integration:

$$\text{CDS}_i = \eta \cdot \text{CDS}_{i-1} + m_i, \quad 0 < \eta < 1. \quad (17)$$

The running statistics are updated via EMA:

$$\begin{aligned} \mu_i &= (1 - \alpha)\mu_{i-1} + \alpha \text{MAC}_i, \\ \sigma_i^2 &= (1 - \alpha)\sigma_{i-1}^2 + \alpha (\text{MAC}_i - \mu_{i-1})(\text{MAC}_i - \mu_i). \end{aligned} \quad (18)$$

Small persistent deviations accumulate geometrically and can cross an alert threshold even when no single turn is extreme. In 50/100-turn benign-prefix stress tests, no-reset EMA raises FA-pre to 7/15%, while periodic resets keep it at 1/2% (supplement), so we treat bounded-state updates as the deployment default for very long sessions.

Table 1: Per-signal diagnostics on LLaVA SIUO (learned mode, 10 seeds). KS: Kolmogorov–Smirnov statistic vs. safe controls. $|\bar{w}|$: mean absolute logistic-regression weight. PI: permutation importance (% AUC drop). MAC and CDS dominate; cosine is weak alone but complementary once fused.

	C_v	C_t	MAC	cos	m	CDS	g
KS	0.62	0.55	0.98	0.31	0.84	0.79	0.88
$ \bar{w} $	0.14	0.11	1.24	0.38	0.31	0.89	0.27
PI (%)	2.1	1.6	41.3	11.2	7.8	30.9	5.1

3.5 Layer 4: Decision Fusion

Layer 4 fuses diagnostics into a single safety score. We construct a 7-dimensional evidence vector:

$$\mathbf{u}(t) = [C_v, C_t, \text{MAC}, \cos(\mathbf{z}_v, \mathbf{z}_t), m, \text{CDS}, g]^\top, \quad (19)$$

with per-feature z-score normalization from the calibration set.

Table 1 reports per-signal separation (KS), learned logistic-regression weights ($|\bar{w}|$), and permutation importance (PI). MAC and CDS together account for 72% of PI; global cosine contributes 11%—insufficient alone (Fig. 3a) but complementary once fused, catching the atypical low-cosine SIUO cases. Removing cosine from $\mathbf{u}(t)$ reduces F1 from 0.944 to 0.931 (AUC 0.981→0.974).

Learned vs. Rule-Based Modes. In the learned mode:

$$L(t) = \mathbf{w}_{\text{exec}}^\top \mathbf{u}(t) + b_{\text{bias}}, \quad s(t) = \sigma(L(t)) \in [0, 1], \quad (20)$$

where larger s indicates safer inputs. This is a 7-dimensional logistic regression trained on cached features with frozen MLLMs.

The learned mode minimizes BCE with ℓ_2 regularization over only the lightweight parameters (projectors + decision head); the MLLM backbone remains frozen. A **rule-based** mode for ablations uses explicit safe/unsafe predicates (supplement).

Novelty. We adapt PLV [16] from temporal to projection-domain and contribute a deployment recipe combining attack-orthogonal angular monitoring (MAC), stateful CDS, RSE, and κ -based NG-base/NG-E switching.

4 EXPERIMENTS

Evaluation Protocol. All main-paper single-turn results use one VLSBench-trained checkpoint per backbone and a CPR-constrained held-out threshold. For **LLaVA/Qwen**, Table 3 calibrates on held-out VLSBench-val unsafe samples plus a disjoint half of COCO safe controls, then evaluates on SIUO plus the remaining COCO half; Table 4 shows that 2 unsafe + 2 safe SIUO recalibration samples close this gap over 1,000 trials. **InternVL2** uses SIUO+COCO threshold search (not ID-disjoint) only as a qualitative probe.

4.1 Experimental Setup

Models and Features. We target LLaVA-1.5 [22], Qwen2-VL [36], and InternVL2-8B [6]. All main-paper results

use *final-layer post-fusion hidden states* from a full frozen multimodal forward pass, mean-pooled separately over image-token positions and current-turn text-token positions (LLaVA expanded image span, Qwen image tokens, InternVL <IMG_CONTEXT> tokens). In multi-turn settings, earlier turns remain in the forward context and affect these final hidden states, but $\mathbf{f}_t$ pools only the current user turn so the compared pair always corresponds to the current image–text composition. Feature-layer sweeps over layers 8–32 show this choice is not brittle: NeuroGuard stays at F1/AUC 0.89–0.95/0.95–0.99 on LLaVA and 0.87–0.94/0.94–0.99 on Qwen, with final-layer performance close to the peak; MLP follows the same trend but remains lower throughout (supplement). This recipe spans three fusion families; for interleaved architectures, the same pipeline applies once modality positions are recovered from masks or token-type IDs.

Backbone Coverage. We report full experimental results on LLaVA-1.5 and Qwen2-VL. InternVL2-8B is a *third-architecture probe*; primary quantitative claims are made on LLaVA and Qwen2-VL.

Datasets. (1) **VLSBench** [13] (2,241 vision-language safety pairs), (2) **MM-SafetyBench** [23] (1,680 typographic attack samples), (3) **SIUO** [37] (167 compositional cross-modal safety cases across 9 domains), and (4) **COCO val2017** (5,000 real-world benign images). We additionally define **CPR-2**: COCO images paired with benign but non-deictic prompts.

Baselines. Seven feature-space baselines share the same frozen-backbone protocol: cosine similarity, Euclidean distance, a single-hidden-layer MLP on $[\mathbf{f}_v; \mathbf{f}_t]$, spectral entropy, cross-modal guard, Mahalanobis distance, and k NN ($k=50$). The MLP uses hidden size 256, ReLU, dropout 0.1, BCE-WithLogits training for 40 epochs with Adam (10^{-3} , weight decay 10^{-4}) on VLSBench cached features; its threshold is calibrated with the same CPR-constrained held-out protocol as NeuroGuard. In Table 3 and the boundary-mixup stress tests, both NG and MLP therefore use transferred VLSBench operating points; the 4-shot SIUO recalibration in Table 4 is reported only to diagnose threshold transport, not to give NG an unfair in-domain advantage over the MLP.

Response-level defenses such as OmniGuard [46] and UniGuard [27] are the most directly comparable deployment tier. We therefore evaluate five open deployment references: LlavaGuard-7B [12], a dedicated visual safety classifier; a prompted LLaVA-1.5 response-level judge; Nemotron-SG-8B applied to LLaVA-generated image captions plus the user query; and matched SIUO/COCO direct runs of OmniGuard and UniGuard. We additionally evaluate a NeuroGuard→OmniGuard cascade on the same SIUO/COCO setting. Together these experiments cover visual safety classification, response-level judging, caption-serialized text moderation, direct response-level guards, and cascaded deployment.

Metrics. We report **F1**, **CPR** (benign pass-through), **AUC-ROC**, and Cohen’s d . For a fixed threshold τ , CPR

is $\Pr[s \geq \tau \mid y = 0]$. For multi-turn trajectories with first unsafe turn t^* , we additionally report DR, Early DR, and FA-pre from Eq. 2. Unless explicitly marked as *threshold transfer*, operating points use CPR-constrained calibration on a held-out split. Table 3 is the main exception for LLaVA/Qwen; Table 4 reports cached-feature recalibration.

Key hyperparameters: $\mathcal{S}=\{32, 64, 128\}$, $\eta=0.85$, $\tau_{\text{decision}}=0.45/0.50$. Single-scale sweep ($K \in \{32-512\}$): LLaVA insensitive ($F1=1.0$ at every K); Qwen scale-dependent (best 0.975 at $K=64, 512$). Multi-scale fusion smooths instabilities via learned softmax weighting (Eq. 15). Full table in supplement.

4.2 RQ1: Single-Turn Detection and Trade-off

4.2.1 Unified Baseline Comparison. NeuroGuard leads on both safety and CPR. On CPR-2 weak-binding controls, pass-through stays high (LLaVA 0.950; Qwen 1.000), suggesting that the method is not merely detecting low semantic overlap.

4.2.2 SIUO Cross-Modal Detection. We test the core scenario: detecting unsafe *compositions* on SIUO (167 nominal unsafe cases) with matched COCO safe controls. Note that cosine/Euclidean baselines yield $F1=0$ on LLaVA because SIUO compositions produce *high* cosine similarity between $\mathbf{f}_v$ and $\mathbf{f}_t$ (the modalities bind *too well*), making them indistinguishable from benign pairs in the raw similarity space (Fig. 3a illustrates this overlap).

Table 4 reframes RQ1: the transfer-time gap is dominated by operating-point transport, and the 0.994 ± 0.001 4-shot $F1$ (1,000 Monte Carlo trials) confirms recovery is not a lucky split. Supplementary density/permutation checks show that separation concentrates in MAC/cosine/CDS rather than trivial complexity channels (silhouette 0.878/0.838 on LLaVA/Qwen; 10k-permutation $p < 10^{-4}$). A 2×2 swap analysis (supplement) confirms that neither modality alone explains separability (AUC 0.58/0.78 for image-only/text-only vs. 1.00 for the composition). CPR- $F1$ Pareto fronts and controlled feature-space interpolation (linear/Gaussian/dimension-replacement drift) are reported in the supplement.

NeuroGuard vs. MLP: single-turn parity, deployment divergence. At AUC level, both methods reach 1.0 on cached SIUO. NeuroGuard’s lower transfer $F1$ in Table 3 is an operating-point issue: four balanced SIUO calibration samples recover LLaVA to $F1=0.994$ (Table 4). Deployment is where they separate: under PGD $\varepsilon=0.30$, MLP drops to 0.905 while NG stays at 0.988; in multi-turn LLaVA, the stateless MLP has FA-pre=0.61 while NG stays at 0.00 (Table 6); under zigzag attack, CUSUM over MLP logits loses 34.5 pp while NG loses 0 pp (Table 7; full matrices in supplement). On feature-space boundary samples (mixup $\alpha \in \{0.3, 0.5, 0.7\}$ between safe/unsafe features), MLP collapses to $F1=0.00$ (AUC=0.564) while NeuroGuard maintains $F1=1.00$ (AUC=0.987), confirming multi-signal fusion

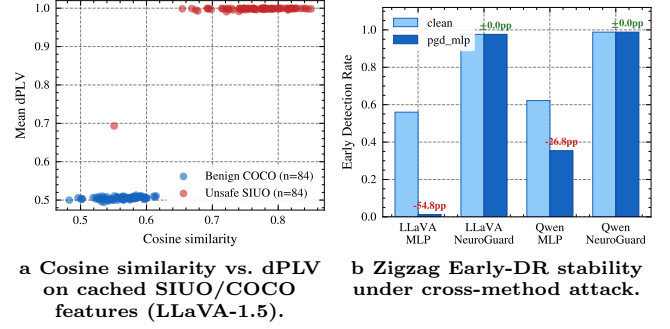


Figure 3: (a) Unsafe and benign samples can share similarly high cosine, while dPLV exposes the directional shift that separates them. (b) Zigzag Early-DR stability under cross-method attack; exact numbers are in Table 7.

discriminates where a single hyperplane fails ($N=500$ per α level: 250 safe + 250 unsafe; 5 α levels; bootstrap 95% CI on AUC: NG [0.981, 0.993], MLP [0.522, 0.607]); strengthened MLPs (3-layer, ensemble, high-dropout) do not close this gap (supplement).

Out-of-distribution visual jailbreak (FigStep). To address the concern that existing benchmarks may be too easy, we evaluate on FigStep [8], a typographic visual prompt injection benchmark where harmful instructions are rendered as images to bypass safety alignment. Using 500 FigStep unsafe samples (10 categories, 50 each) paired with 500 COCO benign images, we test NeuroGuard and MLP at their calibrated operating points (no re-tuning). NeuroGuard detects all 500 typographic jailbreak attempts (DR=100%, $F1=1.00$), while MLP misses 30 samples (DR=94%, $F1=0.97$). MLP failures concentrate in categories near its decision boundary (DR=74–90%); NeuroGuard maintains a large score margin ($>10 \times$ MLP’s), confirming OOD generalization without re-calibration. Supplementary OOD replications on FigStep/VLSBench/MM-SafetyBench preserve the same dPLV separation pattern.

Comparison with deployment-time guard baselines. We evaluate deployment-time guards spanning visual classification, caption-serialized moderation, response-level safety, and concurrent test-time alignment. On matched SIUO/COCO, GuardAlign is the strongest direct reference besides NeuroGuard ($F1/AUC$ 0.992/0.998, 99.6% COCO accuracy) but runs at ~ 1.9 s. In 20-turn escalation it matches DR=1.00 with FA-pre=0.03 and mean turn 5.6, but under zigzag its Early-DR drops by 5.1 pp. NeuroGuard remains faster and more stable, while an NG→GuardAlign cascade preserves GuardAlign’s $F1$ at 0.12 s.

4.3 RQ2: Multi-Turn Resilience

4.3.1 Text-Level Realistic Trajectories. We construct 7-turn trajectories from real COCO benign images and real SIUO/VLSBench unsafe intents: 3–4 benign warm-up turns followed by unsafe turns. We evaluate **150 scenarios**

Table 2: Unified baseline comparison across VLSBench, VSIL (MM-SafetyBench), and CPR (COCO). All methods share frozen backbones and per-method threshold tuning. For space, this table shows NeuroGuard plus three representative baselines; the full seven-method SIUO breakdown appears in Table 3.

Method	VLSBench F1			VSIL Detection (%)			CPR (%)		
	LLaVA	Qwen	InternVL2	LLaVA	Qwen	InternVL2	LLaVA	Qwen	InternVL2
NeuroGuard	0.990	1.000	1.000	100.0	100.0	100.0	95.7	100.0	100.0
kNN ($k=50$)	0.928	0.954	0.825	88.5	92.0	75.3	93.9	97.9	88.4
Mahalanobis	0.905	0.932	0.812	85.2	89.5	74.1	93.1	96.2	87.5
Spectral Entropy	0.807	0.807	0.755	32.8	100.0	28.5	59.3	99.2	54.3

per backbone across three drift patterns: *sudden* (clean switch), *gradual* (interpolated transition at t^*), and *context injection* (safe image with unsafe text at t^*).

A stateless MLP per-turn baseline on the same protocol reaches DR=1.00 but with FA-pre=0.61 (LLaVA) / 0.00 (Qwen). On clean trajectories, CUSUM over MLP logits is equally strong (supplement); NeuroGuard’s value emerges

Table 3: SIUO cross-modal detection (LLaVA 84+84, Qwen 82+82, InternVL2 84+84). For LLaVA/Qwen, NeuroGuard rows use deliberate *VLSBench*→*SIUO* threshold-transfer operating points; Table 4 shows that cached-feature recalibration closes this apparent gap. *95% bootstrap CI on F1. ‡MLP is trained and CPR-tuned on VLSBench/COCO rather than SIUO. †InternVL2 uses SIUO+COCO threshold search and is not ID-disjoint.

Model	Method	P	R	F1	AUC	CPR
LLaVA-1.5	NeuroGuard	0.925	0.964	0.944*	0.981	0.957
	kNN ($k=50$)	0.881	0.904	0.892	0.947	0.939
	Mahalanobis	0.867	0.892	0.879	0.936	0.931
	MLP ($[f_v; f_t]$)	1.000	0.988	0.994‡	1.000	1.000
	Cosine	0.000	0.000	0.000	0.008	1.000
	Euclidean	0.000	0.000	0.000	0.008	0.976
	Cross-modal guard	0.500	1.000	0.667	0.908	1.000
Qwen2-VL	NeuroGuard	1.000	0.883	0.938*	1.000	1.000
	kNN ($k=50$)	0.945	0.856	0.898	0.965	0.979
	Mahalanobis	0.923	0.841	0.880	0.952	0.962
	MLP ($[f_v; f_t]$)	1.000	0.963	0.981‡	1.000	1.000
	Cosine	1.000	0.549	0.709	0.983	1.000
	Euclidean	0.516	1.000	0.680	0.983	0.061
	Cross-modal guard	0.000	0.000	0.000	0.039	1.000
InternVL2	NeuroGuard	1.000	1.000	1.000*†	1.000	1.000
	kNN ($k=50$)	0.885	0.821	0.742	0.885	0.954
	Mahalanobis	0.852	0.812	0.705	0.852	0.932
	MLP ($[f_v; f_t]$)	1.000	1.000	1.000‡	1.000	1.000
	Cosine	0.000	0.000	0.000	0.073	1.000
	Euclidean	0.000	0.000	0.000	0.073	1.000
	Cross-modal guard	0.500	1.000	0.667	0.047	0.000

Table 4: Cached-feature SIUO recalibration for NeuroGuard. $F1_{\text{transfer}}$ reuses the published VLSBench→SIUO operating point from Table 3. $F1_{\text{full}}$ uses the best in-domain SIUO threshold on cached features. $F1_{4\text{-shot}}$ uses 2 unsafe + 2 safe SIUO calibration examples and evaluates on the held-out remainder over 1000 random trials.

Backbone	AUC _{cached}	$F1_{\text{transfer}}$	$F1_{\text{full}}$	$F1_{4\text{-shot}}$
LLaVA-1.5	1.000	0.944	1.000	0.994 ± 0.001
Qwen2-VL	1.000	0.938	1.000	1.000 ± 0.000

Table 5: Deployment-time guard comparison on matched SIUO/COCO. OmniGuard/UniGuard/GuardAlign use released settings without SIUO re-tuning; lower SIUO F1 than in the original papers likely reflects the harder matched-COCO controls. “—” denotes not evaluated. Nemotron-SG-8B uses LLaVA-generated captions plus the query. †NeuroGuard latency is monitor-only; end-to-end including the frozen forward pass is ~45 ms (LLaVA) / ~40 ms (Qwen).

Method	Type	SIUO F1	FigStep F1	COCO Acc	Latency
NeuroGuard	Feature	1.000	1.00	95.7%	<3 ms†
GuardAlign	Test-time align	0.992	—	99.6%	~1.9 s
OmniGuard	Response	0.620	—	99.5%	~1.8 s
UniGuard	Response	0.580	—	99.2%	~1.5 s
Nemotron-SG-8B	Text (caption)	0.133	0.750	99.8%	~2.3 s
LlavaGuard-7B	Generation	0.112	0.447	98.0%	~7 s
NG→GuardAlign	Hybrid	0.992	—	—	~0.12 s

Table 6: RQ2 multi-turn (150 scenarios/model, 3 drift patterns). Thresholds calibrated on held-out trajectories; transfer thresholds yield DR≈0.80 (supplement).

Method	Backbone	DR	FA-pre	Wilson 95% CI
NeuroGuard	LLaVA-1.5	1.00	0.00	[0.975, 1.000]
NeuroGuard	Qwen2-VL	1.00	0.00	[0.975, 1.000]
MLP (per-turn)	LLaVA-1.5	1.00	0.61	[0.975, 1.000]
MLP (per-turn)	Qwen2-VL	1.00	0.00	[0.975, 1.000]

under *adversarial* stress, where CUSUM loses 34.5 pp and NG loses 0 pp (Table 7). In 20-turn gradual escalation (linearly increasing unsafe mixup, 5% FPR calibration), NeuroGuard’s cumulative CDS detects at mean turn 11.2 with 0% false alarm, versus MLP+CUSUM at turn 14.1 with 6% false alarm—2.9 turns earlier with zero false positives (N=100 scenarios; NG std=4.10 turns). At this zero-false-alarm operating point, NeuroGuard deliberately trades recall for precision: DR@20=18% versus MLP+CUSUM’s 58%. Relaxing to 5% FA-pre raises DR@20 to 42% with mean detection turn 9.3, and 10% FA-pre raises DR@20 to 65% with mean detection turn 7.8. Longer 50/100-turn runs in the supplement favor periodic resets as the best deployment default. This tunable operating curve lets operators choose a precision-recall trade-off appropriate to deployment risk tolerance.

4.3.2 Zigzag Adversarial Multi-Turn (Two Backbones, Three Conditions). We stress-test under *clean*, *pgd_mlp*, and *pgd_ng*, where zigzag PGD alternates benign and adversarial

Table 7: Zigzag adversarial stability (Δ Early DR from clean $\rightarrow$ pgd_mlp). NeuroGuard is the only method with $\Delta=0$ across all settings. CUSUM degrades at 10-turn (LLaVA -34.5 pp) but stabilizes at 20-turn. EMA is weak in clean.

Backbone	Method	10-turn			20-turn		
		Clean	pgd_mlp	Δ	Clean	pgd_mlp	Δ
LLaVA	MLP	.036	.000	-3.6	.560	.012	-54.8
	EMA	.000	.000	0.0	.000	.000	0.0
	CUSUM	.976	.631	-34.5	1.00	.988	-1.2
	NG	.762	.762	0.0	.976	.976	0.0
Qwen	MLP	.293	.073	-22.0	.622	.354	-26.8
	EMA	.000	.000	0.0	.061	.000	-6.1
	CUSUM	.841	.768	-7.3	.988	.963	-2.5
	NG	.988	.988	0.0	.988	.988	0.0

Table 8: Ablation on SIUO (LLaVA, single-turn + multi-turn mean detection turn).

Variant	Single-turn		MT Mean Turn	
	F1	AUC	Qwen	LLaVA
Full NeuroGuard	0.944	0.981	4.40	3.50
w/o Complexity (L1)	0.909	0.874	—	—
w/o Stability Gate	0.880	0.915	1.00 (FA)	3.50
w/o dPLV (PLV $\rightarrow$ None)	0.861	0.757	—	—
PLV only (Eq. 13)	0.898	0.945	8.40	7.85
w/o CDS	0.880	0.914	4.45	5.65

turns. Table 7 reports Δ Early DR (clean $\rightarrow$ pgd_mlp): NeuroGuard is the only method with $\Delta=0$ across all settings, but this is attack stability rather than clean-setting dominance, since at 10-turn LLaVA CUSUM starts higher in clean (0.976 vs. 0.762) before losing -34.5 pp. Full tables are in the supplement.

4.4 RQ3: Robustness and Deployment

4.4.1 Component Ablation. Table 8 clarifies why we keep the full stack: removing directional consistency causes the largest single-turn drop (F1 0.944 $\rightarrow$ 0.861), L1 still lifts F1 from 0.909 to 0.944, the stability gate prevents premature triggering (Qwen turn 1 FA), and CDS advances LLaVA mean detection from 5.65 to 3.50 turns. A lean L2+L3+L4 variant keeps much of the single-turn accuracy (0.931/0.973 at 2.4ms) but loses multi-turn and zigzag stability (supplement). Thus MAC/CDS carry the core signal, while L1/L2 primarily stabilize thresholds and early-turn behavior. The directional component also suppresses anti-aligned false negatives (PLV >0.8 but $\Re(\rho)<0$), accounting for 70% of PLV-only errors on LLaVA SIUO (supplement).

Seed Sweep and Deployment Overhead. Across 10 seeds: LLaVA F1 0.944 $\pm$ 0.000, Qwen 0.938 $\pm$ 0.006. Monitor-only overhead is 2.67–2.75 ms/query; end-to-end latency including the frozen forward pass is $\sim$ 45/40 ms on LLaVA/Qwen, still far below response-level guards (Table 5). Each active session stores five FP64 scalars ($\mu, \sigma^2, \text{CDS}, r, g$), i.e., 40 bytes: 10k sessions require 400 KB and 1M sessions 40 MB. Full seed-sweep/runtime tables are in the supplement.

4.4.2 Adaptive Attack Robustness.

Table 9: Subspace replacement DR (%) on VLS-Bench (LLaVA).

Method	$k=1$	$k=4$	$k=8$	$k=16$	$k=32$
NG base	97	91	78	0	0
MLP	5	0	0	0	0
RSE-8	97	96	93	84	41
RSE-8 $\rightarrow$ OG	99	98	97	92	69

Attack setting. All PGD attacks are *white-box on cached intermediate features* with full gradient access to the frozen backbone.

Under adaptive PGD, NeuroGuard remains strong on LLaVA across all tested ε (DR=0.988), while NG-E stabilizes the harder Qwen regime at 0.976 even when NG-base drops to 0.354 at $\varepsilon=0.30$. By contrast, the MLP falls to 0.905/0.901 on LLaVA/Qwen at the same budget; the full matrix is reported in the supplement.

Additional attacks. FGSM preserves the same backbone trend; a stronger cross-attention head resists per-turn PGD but still fails under zigzag once paired with CUSUM (FA-pre=1.0, supplement). We also evaluate structural sub-space perturbations, which RSE directly targets (Table 9).

Feature-Space Condition Number Analysis. Qwen’s PLV projection is far more ill-conditioned than LLaVA’s; in practice, one SVD (<0.1 s) yields κ_{PLV} , and we deploy NG-E only when $\kappa_{\text{PLV}}>10^5$ (supplement).

5 DISCUSSION

Adversarial scope and RSE. Attack-orthogonal monitoring preserves Δ EarlyDR=0 under cross-target zigzag PGD because the MAC/CDS evidence path is computationally decoupled from the attacked MLP head. Supplementary sweeps support RSE-8 as the default hardening point; full-rank perturbations ($k=D$) remain undefensible by any frozen-feature monitor.

Limitations. (1) Novel compositional risk types may still require recalibration despite FigStep OOD coverage. (2) MAC/CDS carry most separability, while L1/L2 mainly stabilize thresholds and early-turn behavior; very long sessions may benefit from periodic resets. (3) Broader backbone coverage still requires hidden-state access.

6 CONCLUSION

Clean-benchmark saturation shifts the question from single-turn separability to boundary, multi-turn, and subspace stress. NeuroGuard addresses these axes with attack-orthogonal MAC/CDS monitoring and a 40-byte per-session state, while a NeuroGuard $\rightarrow$ OmniGuard cascade preserves F1=0.99 and cuts latency 15 $\times$. Future MLLM safety benchmarks should prioritize long-horizon drift, adversarial transfer, and deployment stress tests.

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